

Prove or disprove saddle-escaping for subgradient methods in robust matrix sensing

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Website #22 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

Fix the robust Burer–Monteiro ℓ_1 objective in the matrix sensing model

$$f(W) = \frac{1}{m} \sum_{i=1}^m |y_i - \langle A_i, WW^\top \rangle|$$

in the symmetric PSD case, with search rank $k \geq r$, and let $W_\star$ be a *true* factorization of $X_\star$, i.e. $W_\star W_\star^\top = X_\star$. The question is whether the deterministic subgradient iteration

$$W_{t+1} = W_t - \eta_t g_t, \quad g_t \in \partial f(W_t),$$

must almost surely avoid convergence to $W_\star$ whenever $W_\star$ is a strict saddle (in the sense that there exists a direction V and $c > 0$ with $f(W_\star + \gamma V) \leq f(W_\star) - c\gamma^2$ for all sufficiently small γ), or whether, despite this strict-saddle geometry, there can exist an initialization rule and a legitimate deterministic subgradient selector for which $W_t \rightarrow W_\star$ with positive probability.

The negation of the universal saddle-avoidance claim is:

$$\exists \text{ an instance, } \exists \text{ a deterministic selector } g(\cdot), \exists (\eta_t), \exists \nu$$

such that $W_\star$ is a strict saddle true solution and

$$\mathbb{P}_{W_0 \sim \nu}(W_t \rightarrow W_\star) > 0.$$

I focus on the hardest operational reading: the sensing/noise instance is fixed, $W_\star$ is already known to be a strict saddle of that fixed objective, and the probability is over the initialization law (the algorithm itself being deterministic once the selector is fixed).

Alternative readings considered

Two nearby readings are possible.

- Probability could be taken jointly over the Gaussian measurement draw and the initialization. I do *not* settle that blanket Gaussian statement.
- One could ask for attraction from an absolutely continuous initialization law in the ambient factor space. I do not prove that either.

What I *do* prove is that strict-saddle geometry alone does *not* force deterministic subgradient dynamics to avoid the saddle: there is a concrete invariant-manifold mechanism making a true strict saddle attracting.

Related literature found

- **Ma–Fattahi (2025)** — PARTIAL. This is the closest geometric source: it shows that in robust low-rank matrix recovery, true solutions need not be local optima and can instead be strict saddles under moderate assumptions. It addresses landscape, not deterministic saddle-escape dynamics. (pubsonline.informs.org)
- **Ma–Fattahi (2023, JMLR)** — PARTIAL. This paper proves that in the Gaussian measurement model, SubGM with small initialization and exponentially decaying stepsizes converges to the true matrix in the over-parameterized noisy setting; however, the theorem is phrased at the matrix level and does not directly prove convergence to a prescribed factor $W_\star$. (jmlr.org)
- **Ding–Jiang–Chen–Qu–Zhu (NeurIPS 2021)** — PARTIAL. They prove that, under RDPP, overspecified-rank subgradient descent with diminishing stepsizes converges to the exact low-rank solution, and they report that RDPP holds generically for Gaussian measurements with sparse corruptions. This is very suggestive, but again not phrased as factor-level convergence to a designated strict saddle. (proceedings.neurips.cc)
- **Li–Zhu–So–Vidal (2020)** — TOOL. This exact-rank robust recovery paper shows sharpness and weak convexity of the ℓ_1 -factorized objective, leading to local SubGM convergence when suitably initialized. I checked it as a possible ingredient for a Gaussian exact-rank reduction argument, but I ultimately kept the proof below self-contained. (epubs.siam.org)
- **Davis–Drusvyatskiy (2019)** — BACKGROUND. This paper is conceptually crucial: it emphasizes that mere quadratic decrease/negative curvature is insufficient for saddle avoidance in nonsmooth optimization. That observation strongly suggested looking for an invariant-manifold counterexample here. (sites.math.washington.edu)
- **Bianchi–Hachem–Schechtman (2024)** — BACKGROUND. They prove that *stochastic* subgradient descent avoids *active* strict saddles under additional stratification/angle assumptions. I checked this carefully because it looked like a possible direct tool; it does not apply to the example I construct, and in fact the key active-manifold sharpness fails there. (pubsonline.informs.org)

Proof sketch

The mechanism is simple and, I believe, the main conceptual point.

In the symmetric Burer–Monteiro parametrization $X = WW^\top$, every Clarke subgradient has the form

$$g(W) = G(W)W$$

for some square matrix $G(W)$ built from the sensing matrices and residual signs. Therefore any zero redundant column of W remains zero forever under the update $W^+ = W - \eta g(W)$. So the over-parameterized dynamics have an *exact-rank invariant manifold*

$$\mathcal{M} := \{[U, 0]\}.$$

If a true factor $U_\star$ is attracting for the reduced exact-rank dynamics on $\mathcal{M}$, then the lifted point $[U_\star, 0]$ is attracting for the full over-parameterized dynamics *along that invariant manifold*, regardless of what happens transversely. Hence, if the full objective happens to make $[U_\star, 0]$ a

strict saddle by creating a quadratic descent direction in a redundant column, the deterministic subgradient method can still converge to that strict saddle whenever it is initialized on $\mathcal{M}$.

I then instantiate this mechanism in the smallest nontrivial symmetric example I could make work: $d = 2$, true rank $r = 1$, search rank $k = 2$, and $m = 3$ measurements with one outlier ($p = 1/3$). On the invariant manifold $\mathcal{M}$, the iteration reduces to a one-dimensional recursion

$$u_{t+1} = u_t \left(1 \pm \frac{1}{t+2} \right),$$

whose convergence to $u = 1$ is proved by passing to $z_t = \log u_t$ and showing that sign changes force the overshoot to vanish.

So the final answer is: *strict saddles can indeed be attracting for deterministic subgradient dynamics in this robust BM- ℓ_1 setting.* What remains open in my manuscript is the blanket version for generic Gaussian instances and generic ambient random initialization laws.

Main result

Theorem 1. *In the symmetric PSD robust matrix sensing BM- ℓ_1 model, strict-saddle geometry does not force deterministic subgradient dynamics to avoid the true solution.*

More precisely:

- (a) (Invariant-manifold lifting) *For any sensing instance and any deterministic residual-sign selector, the manifold of zero redundant columns*

$$\mathcal{M}_r := \left\{ W = [U, 0] \in \mathbb{R}^{d \times k} : U \in \mathbb{R}^{d \times r} \right\}$$

is invariant under deterministic subgradient descent.

Hence any attracting true solution $U_\star$ of the reduced exact-rank dynamics on $\mathcal{M}_r$ lifts to an attracting point $[U_\star, 0]$ of the full over-parameterized dynamics on $\mathcal{M}_r$. In particular, if $[U_\star, 0]$ is a strict saddle of the full objective, then deterministic subgradient descent can converge to a strict saddle.

- (b) (Explicit attracting strict saddle) *There exists a concrete robust matrix sensing instance with*

$$d = 2, \quad r = 1, \quad k = 2, \quad m = 3, \quad p = 1/3 < 1/2,$$

a true solution

$$W_\star = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

a deterministic selector $g(W) \in \partial f(W)$, and the standard diminishing stepsize

$$\eta_t = \frac{3}{2(t+2)},$$

such that:

- (i) *$W_\star$ is Clarke critical and is a strict saddle of f ;*
- (ii) *for every initialization*

$$W_0 = \begin{pmatrix} u_0 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{with } u_0 > 0,$$

the iterates satisfy $W_t \rightarrow W_\star$.

Consequently, if $u_0 \sim \text{Unif}(0, 1)$, then

$$\mathbb{P}(W_t \rightarrow W_\star) = 1.$$

Therefore the universal claim “deterministic subgradient iterates almost surely avoid strict-saddle true solutions” is false at the level of general BM- ℓ_1 robust matrix sensing dynamics.

Proof

I write E_{ij} for the 2×2 standard basis matrix with a 1 in entry (i, j) , and I use

$$\text{Sign}(t) = \begin{cases} \{1\}, & t > 0, \\ [-1, 1], & t = 0, \\ \{-1\}, & t < 0, \end{cases} \quad \text{sgn}_0(t) = \begin{cases} 1, & t > 0, \\ 0, & t = 0, \\ -1, & t < 0. \end{cases}$$

Lemma 1 (Form of the subgradient in the symmetric PSD model). *Let*

$$f(W) = \frac{1}{m} \sum_{i=1}^m |y_i - \langle A_i, WW^\top \rangle|,$$

where each A_i is symmetric. Then

$$\partial f(W) = \left\{ -\frac{2}{m} \left(\sum_{i=1}^m s_i A_i \right) W : s_i \in \text{Sign}(y_i - \langle A_i, WW^\top \rangle) \right\}.$$

Proof. Define

$$r_i(W) := y_i - \langle A_i, WW^\top \rangle.$$

Since A_i is symmetric,

$$\langle A_i, WW^\top \rangle = \text{trace}(A_i WW^\top) = \text{trace}(W^\top A_i W),$$

and differentiating yields

$$\nabla \langle A_i, WW^\top \rangle = 2A_i W.$$

Hence

$$\nabla r_i(W) = -2A_i W.$$

Now $W \mapsto |r_i(W)|$ is the composition of a C^1 scalar map r_i and the absolute value. Its Clarke subdifferential is

$$\partial |r_i(W)| = \{\sigma \nabla r_i(W) : \sigma \in \text{Sign}(r_i(W))\} = \{-2\sigma A_i W : \sigma \in \text{Sign}(r_i(W))\}.$$

Summing over i and dividing by m gives the formula. $\square$

Lemma 2 (Invariant redundant-column manifold). *Fix any deterministic selector $s_i(W) \in \text{Sign}(y_i - \langle A_i, WW^\top \rangle)$, and let*

$$g(W) := -\frac{2}{m} \left(\sum_{i=1}^m s_i(W) A_i \right) W.$$

Partition $W = [U, Z]$ with $U \in \mathbb{R}^{d \times r}$ and $Z \in \mathbb{R}^{d \times (k-r)}$. Then under

$$W^+ = W - \eta g(W),$$

one has

$$Z^+ = \left(I + \frac{2\eta}{m} \sum_{i=1}^m s_i(W) A_i \right) Z.$$

In particular, if $Z = 0$, then $Z^+ = 0$. Therefore

$$\mathcal{M}_r = \{[U, 0]\}$$

is invariant.

Proof. Write

$$M(W) := \frac{2}{m} \sum_{i=1}^m s_i(W) A_i.$$

Then $g(W) = -M(W)W$, so

$$W^+ = W + \eta M(W)W = (I + \eta M(W))W.$$

Since $W = [U, Z]$,

$$W^+ = [(I + \eta M(W))U, (I + \eta M(W))Z].$$

The claimed formula for Z^+ follows immediately. If $Z = 0$, then $Z^+ = 0$. □

Lemma 3 (A scalar recursion converging to 1). *Let*

$$u_{t+1} = \begin{cases} u_t \left(1 + \frac{1}{t+2} \right), & 0 < u_t < 1, \\ u_t, & u_t = 1, \\ u_t \left(1 - \frac{1}{t+2} \right), & u_t > 1. \end{cases}$$

Then for every $u_0 > 0$, one has $u_t \rightarrow 1$.

Proof. Set $z_t := \log u_t$. Since $u_t > 0$ for all t , this is well-defined.

If $z_t < 0$, then $u_t < 1$, hence

$$z_{t+1} = z_t + \log \left(1 + \frac{1}{t+2} \right) = z_t + a_t, \quad a_t := \log \frac{t+3}{t+2} > 0.$$

If $z_t > 0$, then $u_t > 1$, hence

$$z_{t+1} = z_t + \log \left(1 - \frac{1}{t+2} \right) = z_t - b_t, \quad b_t := \log \frac{t+2}{t+1} > 0.$$

If $z_t = 0$, then $u_t = 1$ and $z_{t+1} = 0$.

Now

$$\sum_{t=0}^{T-1} a_t = \log \frac{T+2}{2} \xrightarrow{T \rightarrow \infty} \infty, \quad \sum_{t=0}^{T-1} b_t = \log(T+1) \xrightarrow{T \rightarrow \infty} \infty.$$

Therefore z_t cannot stay eventually negative, because then it would be increased by an unbounded total amount; similarly it cannot stay eventually positive. So either z_t hits 0 in finite time and then stays there, or else it changes sign infinitely many times.

Consider a sign change between times t and $t+1$. If $z_t < 0 \leq z_{t+1}$, then

$$0 \leq z_{t+1} = z_t + a_t \leq a_t.$$

If $z_t > 0 \geq z_{t+1}$, then

$$0 \leq -z_{t+1} = -z_t + b_t \leq b_t.$$

Hence every overshoot after a sign change satisfies

$$|z_{t+1}| \leq \max\{a_t, b_t\} \xrightarrow[t \rightarrow \infty]{} 0.$$

Between two consecutive sign changes, the sign of z_t is constant; in that interval, $|z_t|$ decreases monotonically, because we add a positive amount when $z_t < 0$ and subtract a positive amount when $z_t > 0$. Therefore the maximum of $|z_t|$ after the N -th sign change is bounded by the overshoot at that sign change, which tends to 0. Thus $z_t \rightarrow 0$, i.e.

$$u_t = e^{z_t} \rightarrow 1.$$

□

Proof of Theorem. Part (a) is exactly Lemma 2. So I only need to construct the explicit instance for part (b).

Step 1: the instance. Take

$$X_\star = E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad W_\star = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad W_\star W_\star^\top = X_\star.$$

Let

$$A_1 = E_{11}, \quad A_2 = E_{22}, \quad A_3 = E_{12} + E_{21}.$$

Take measurements

$$y_1 = \langle A_1, X_\star \rangle = 1, \quad y_2 = \langle A_2, X_\star \rangle + 1 = 1, \quad y_3 = \langle A_3, X_\star \rangle = 0.$$

Thus exactly one of the three samples is corrupted, namely index 2, so $p = 1/3 < 1/2$. The corrupted noise is deterministic and equal to 1, which satisfies the required “mass away from zero” condition, e.g.

$$\mathbb{P}(|\epsilon_2| \geq 1) = 1.$$

For a generic

$$W = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

we compute

$$\langle A_1, WW^\top \rangle = a^2 + b^2, \quad \langle A_2, WW^\top \rangle = c^2 + d^2, \quad \langle A_3, WW^\top \rangle = 2(ac + bd).$$

Hence

$$f(W) = \frac{1}{3} \left(|1 - a^2 - b^2| + |1 - c^2 - d^2| + 2|ac + bd| \right).$$

Step 2: a deterministic selector. Define residuals

$$r_i(W) := y_i - \langle A_i, WW^\top \rangle, \quad i = 1, 2, 3,$$

and choose the deterministic selector

$$s_i(W) := \text{sgn}_0(r_i(W)).$$

By Lemma 1,

$$g(W) := -\frac{2}{3}(s_1(W)A_1 + s_2(W)A_2 + s_3(W)A_3)W \in \partial f(W).$$

Step 3: $W_\star$ is Clarke critical. At $W_\star$,

$$r_1(W_\star) = 0, \quad r_2(W_\star) = 1, \quad r_3(W_\star) = 0.$$

Therefore

$$s_1(W_\star) = 0, \quad s_2(W_\star) = 1, \quad s_3(W_\star) = 0,$$

and so

$$g(W_\star) = -\frac{2}{3}A_2W_\star = 0,$$

because $A_2W_\star = 0$. Since $g(W_\star) \in \partial f(W_\star)$, we have $0 \in \partial f(W_\star)$: $W_\star$ is Clarke critical.

Step 4: $W_\star$ is a strict saddle. Take the direction

$$V := E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then

$$W_\star + \gamma V = \begin{pmatrix} 1 & 0 \\ 0 & \gamma \end{pmatrix}, \quad (W_\star + \gamma V)(W_\star + \gamma V)^\top = \begin{pmatrix} 1 & 0 \\ 0 & \gamma^2 \end{pmatrix}.$$

For $|\gamma| < 1$,

$$f(W_\star + \gamma V) = \frac{1}{3}(|1 - 1| + |1 - \gamma^2| + 0) = \frac{1}{3}(1 - \gamma^2).$$

Since

$$f(W_\star) = \frac{1}{3},$$

we obtain

$$f(W_\star + \gamma V) - f(W_\star) = -\frac{1}{3}\gamma^2.$$

Thus $W_\star$ admits a quadratic decrease direction and is a strict saddle in the sense asked in the problem.

Step 5: dynamics on the invariant manifold. Fix any initialization

$$W_0 = \begin{pmatrix} u_0 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{with } u_0 > 0.$$

By Lemma 2, every iterate has the form

$$W_t = \begin{pmatrix} u_t & 0 \\ 0 & 0 \end{pmatrix}$$

for some scalar $u_t > 0$.

At such a point,

$$r_1(W_t) = 1 - u_t^2, \quad r_2(W_t) = 1, \quad r_3(W_t) = 0.$$

Hence

$$s_1(W_t) = \text{sgn}_0(1 - u_t^2), \quad s_2(W_t) = 1, \quad s_3(W_t) = 0.$$

Since

$$A_2 W_t = 0, \quad A_3 W_t = 0,$$

the selected subgradient becomes

$$g(W_t) = -\frac{2}{3} \text{sgn}_0(1 - u_t^2) \begin{pmatrix} u_t & 0 \\ 0 & 0 \end{pmatrix}.$$

With the stepsize

$$\eta_t = \frac{3}{2(t+2)},$$

the update

$$W_{t+1} = W_t - \eta_t g(W_t)$$

reduces to

$$u_{t+1} = \begin{cases} u_t \left(1 + \frac{1}{t+2}\right), & 0 < u_t < 1, \\ u_t, & u_t = 1, \\ u_t \left(1 - \frac{1}{t+2}\right), & u_t > 1. \end{cases}$$

By Lemma 3, $u_t \rightarrow 1$. Therefore

$$W_t \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = W_\star.$$

If now $u_0 \sim \text{Unif}(0, 1)$, then $u_0 > 0$ almost surely, so the above deterministic convergence gives

$$\mathbb{P}(W_t \rightarrow W_\star) = 1.$$

This proves part (b), hence the theorem. □

Gap to the full statement

The full prompt asked about the *Gaussian* sensing model and about whether subgradient dynamics almost surely avoid strict-saddle true solutions there. My proof above does *not* settle that blanket Gaussian statement.

What I have proved is stronger in one direction and weaker in another:

- Stronger conceptually: it gives a precise mechanism showing that strict-saddle geometry *alone* does not imply avoidance for deterministic subgradient methods. A true strict saddle can be attracting because the dynamics may live on an invariant exact-rank manifold.
- Weaker statistically: my explicit construction uses deterministic sensing matrices, not i.i.d. Gaussian ones.

What remains open in this manuscript is the following sharper Gaussian version:

Given an i.i.d. Gaussian sensing instance for which a true factor $W_\star$ is a strict saddle, and given an ambient initialization law that is absolutely continuous near the origin, does deterministic SubGM satisfy $\mathbb{P}(W_t \rightarrow W_\star) = 0$, or can that probability be positive?

The literature does, however, push in the same direction as the mechanism above. Ma–Fattahi (2025) show that true solutions can indeed be strict saddles in robust low-rank recovery; Ma–Fattahi (2023) show that, with Gaussian measurements and small initialization, SubGM still converges to the ground-truth matrix in the over-parameterized noisy regime; and Ding et al. (2021) prove exact low-rank recovery under RDPP, with RDPP holding generically for Gaussian sparse corruptions. So the available Gaussian theory already says that “true strict saddle” and “successful subgradient recovery” coexist; what is still missing is a clean factor-level theorem for convergence to a *prescribed* $W_\star$ from a generic ambient initialization law. (pubsonline.informs.org)

A final important nuance is that active-strict-saddle escape theorems do *not* contradict my example. In my construction, the subgradient norm is *not* bounded away from zero off the attracting manifold: for

$$\widetilde{W}_\gamma = \begin{pmatrix} 1 & 0 \\ 0 & \gamma \end{pmatrix},$$

one has

$$\text{dist}(0, \partial f(\widetilde{W}_\gamma)) \leq \left\| -\frac{2}{3} A_2 \widetilde{W}_\gamma \right\|_F = \frac{2}{3} |\gamma| \rightarrow 0.$$

So the sharp normal-growth condition underlying the “active strict saddle” theory fails here, exactly as Davis–Drusvyatskiy warned can happen in nonsmooth problems. (sites.math.washington.edu)

Self red-team

Counterexample attempts to my own theorem.

- I first tried $d = 1$. That cannot work. Indeed in the symmetric $d = 1$ case the objective is $f(w) = F(w^2)$ with

$$F(x) = \frac{1}{m} \sum_{i=1}^m |y_i - a_i x|,$$

and F is convex in x . If $w_\star \neq 0$ is Clarke critical, then $0 \in \partial f(w_\star) = 2w_\star \partial F(w_\star^2)$, hence $0 \in \partial F(w_\star^2)$, so $w_\star^2$ minimizes the convex function F . Therefore $w_\star$ cannot be a strict saddle. So $d = 2$ really is the first place the phenomenon can occur.

- I tried $m = 2$ with one corrupted sample, but then $p = 1/2$, which sits on the usual boundary and is less faithful to the stated regime. Adding a third zero-residual measurement gives $p = 1/3$ and keeps the construction clean.
- I tried to enlarge the basin to an open ambient neighborhood. That failed: off the invariant exact-rank manifold, the negative-curvature direction is genuinely active, so the explicit example does not have an open Euclidean basin around $W_\star$.

Three likely referee objections and responses.

- *Objection 1: You used a special subgradient choice at zero residuals.* Response: yes, deliberately. A deterministic subgradient method is not defined until one specifies a selector $g(W) \in \partial f(W)$. The choice $\text{sgn}_0(0) = 0$ is legitimate and standard. Since the claim being tested is a universal avoidance claim for deterministic subgradient dynamics, one valid deterministic selector suffices to refute it.
- *Objection 2: Your random initialization is singular (supported on a lower-dimensional manifold).* Response: correct. This is why the verdict is **PARTIAL**, not **SOLVED**. The theorem proves that strict saddles can attract with positive probability; it does not prove attraction from an absolutely continuous ambient initialization law.
- *Objection 3: Your explicit sensing operator is not Gaussian.* Response: also correct. The explicit theorem is an existence result inside the same BM- ℓ_1 family. The remaining gap is exactly the Gaussian genericity issue stated in the previous section.

Boundary checks.

- If $u_0 = 1$, the algorithm is already at $W_\star$ and stays there.
- If $u_0 = 0$, the iterate stays at 0, so convergence to $W_\star$ fails; hence I explicitly require $u_0 > 0$.
- If $u_0 < 0$, the same scalar analysis shows convergence to -1 , i.e. to the different true solution $\begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}$.
- The strict-saddle inequality is local: I used $|\gamma| < 1$, precisely where $|1 - \gamma^2| = 1 - \gamma^2$.

Ideas tried

- **Approach 1: Prove the literal Gaussian statement by combining existing theorems.** Idea: merge the strict-saddle geometry result of Ma–Fattahi with Gaussian SubGM convergence results (Ma–Fattahi 2023, Ding et al. 2021). Obstruction: the available convergence theorems are primarily matrix-level or solution-set-level, whereas the prompt asks for convergence to a *particular* factor $W_\star$. Factor nonuniqueness and gauge freedom become the sticking point. Natural next step: impose a gauge condition or work on a quotient manifold, then lift the matrix-level result back to a distinguished representative.
- **Approach 2: Perturb my explicit deterministic example to an open set of Gaussian sensing instances.** Idea: use the full support of the Gaussian law and a structural-stability argument. Obstruction: the clean one-dimensional reduction in the example relies on an exact invariant axis/eigenspace structure. Generic perturbations destroy that exact reduction, and I did not complete a robust Lyapunov analysis for the perturbed two-dimensional reduced dynamics. Natural next step: prove a cone-stability result for the reduced exact-rank dynamics under small perturbations of the sensing matrices.
- **Approach 3: Find a smaller $d = 1$ counterexample.** Idea: make the phenomenon scalar so that the iteration can be solved exactly. Obstruction: in $d = 1$, after collapsing to the matrix variable x , the objective is convex in x , and a nonzero Clarke-critical true solution cannot be a strict saddle. Natural next step: none; this route is genuinely blocked.
- **Approach 4: Seek an open ambient basin, not just an invariant manifold.** Idea: show positive-probability convergence for absolutely continuous random initialization in the

explicit $d = 2$ example. Obstruction: the redundant-column direction is the very direction of quadratic decrease, and the subgradient dynamics do not damp it; off the exact-rank manifold the iterate is pushed away rather than pulled back. Natural next step: engineer a different sensing design where the transverse nonsmooth term dominates the negative curvature and creates a cone of attraction, closer in spirit to Davis–Drusvyatskiy’s examples.

Invoked results

No external theorem is used as a step in the proof above. Every mathematical step in the proof is derived directly in this manuscript. The web sources are used only for orientation, context, and to delineate the remaining gap to the full Gaussian statement.

Self-confidence

3 — I am confident in the explicit counterexample, the invariant-manifold argument, and the convergence proof; the only substantive uncertainty is the unresolved extension to generic Gaussian sensing and generic ambient initialization.

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